

SKEMA

Derivatives and Fixed Income
PGE M1 Fall 2026

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Session 12: Monte-Carlo Simulation & Loose Ends

Recall: Black-Scholes Model Assumptions



BS assumes

- Stock prices & returns are log-normally distributed
  - Prices follow a geometric Brownian motion
- Volatility of returns is constant
- There is continuous trading
- There are no transaction costs, no taxes
- There are no dividends paid on the underlying
- There are no riskless arbitrage opportunities
- Short-selling is permitted
- r is fixed
- For European options

Black-Scholes vs the Real World



In reality, there is a problem with every assumption!

- Markets close every evening, on weekends and holidays
- Volatility spikes and there is a term-structure of interest rates
- Stock prices have jumps and returns have a left-tail distribution
- Most traded options are American style
- Transaction costs add up

Simulations



The appeal of Black-Scholes is the closed-form expressions for the value of European calls & puts, which allow us to compute option prices nearly instantly.

A competing approach is to rely on numerical simulations that allow us to relax many of the assumptions under BS

1. Simulate many random sample paths
2. Compute the expected payoff for each path
3. Average across paths to see what happens “on average”

Geometric Brownian Motion (**)

Assume time is continuous ($\Delta t \to 0$), stock prices follow a GBM

$$\frac{\mathrm{d}S_t}{S_t} = \mu \mathrm{d}t + \sigma \mathrm{d}Z_t$$

where W_t is a standard Brownian motion and $Z_0 = 0,\ \mathrm{d}Z_t \sim N(0, \mathrm{d}t)$.

To interpret,

- μ is the instantaneous expected rate of return, or the average growth rate in the expected price.
- σ is the volatility of returns, or the size of "randomness" in the price path.

Geometric Brownian Motion

The solution to the stochastic differential equation for a GBM is

$$S_T = S_0 e^{(u - \frac{1}{2}\sigma^2)T + \sigma Z_T}$$

Under the risk-neutral measure $\mathbb{Q}$, which we use to compute option prices, the price path follows

$$S_T^{\mathbb{Q}} = S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma Z_T}$$

- In a risk-neutral world, investors do not demand compensation for risk, so the required rate of return on any asset is simply the risk-free rate instead of μ.
- The $-\frac{1}{2}\sigma^2$ term is a correction term due to how the randomness affects the average growth rate.

Monte Carlo Simulation (**)

To compare the Black-Scholes price of a European call versus the price computed via Monte-Carlo, we will do the following:
Let $S_0 = 330, r = 4\%, K = 350, \sigma = 0.25, T = 3$ months.

For a single sample path

i From date 0 until $T = 3$ months, draw a random vector of values from a normal distribution with mean 0 and variance $\mathrm{d}t$ (in practice, set $\mathrm{d}t = 1/252$).

ii Update the terminal stock price $S_T^{\mathbb{Q}}$ according to $S_T^{\mathbb{Q}} = S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma Z_T}$, where $Z_T = \int_0^T \mathrm{d}Z_t$.

iii Calculate the payoff of the call as $\max(S_T - K, 0)$.

Repeat steps (i)-(iii) many, many times, and take the average discounted payoff across all the simulations as the price of the call.

Monte Carlo Call Pricing via Python (**)



```
import numpy as np

def MC_call(S0, K, T, r, sigma, n_paths=100000):
    Z = np.random.standard_normal(n_paths)
    ST = S0 * np.exp((r - 0.5*sigma**2)*T + sigma*np.sqrt(T)*Z)
    payoffs = np.maximum(ST - K, 0)
    discounted = np.exp(-r*T) * payoffs
    price = np.mean(discounted)
    se = np.std(discounted) / np.sqrt(n_paths)
    return price, se

price, se = MC_call(330, 350, 0.25, 0.04, 0.25,1000)
print(f"Monte Carlo Call Price: {price:.2f} +/- {1.96*se:.2f}
    (95% CI) USD")
```

Black-Scholes Comparison via Python (**)

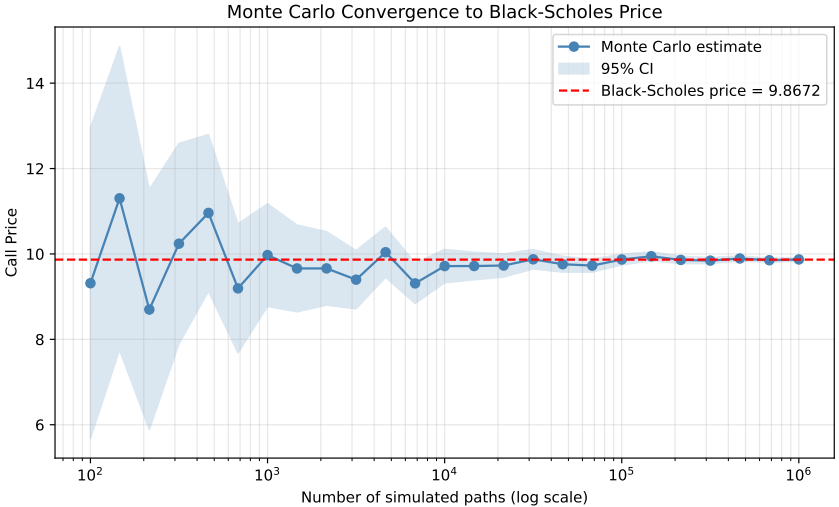


```
from scipy.stats import norm

def bs_call(S0, K, T, r, sigma):
    d1 = (np.log(S0/K) + (r + 0.5*sigma**2)*T) / (sigma*np.sqrt
        (T))
    d2 = d1 - sigma*np.sqrt(T)
    return S0*norm.cdf(d1) - K*np.exp(-r*T)*norm.cdf(d2)

bs_price = bs_call(330, 350, 0.25, 0.04, 0.25)
print(f"Black-Scholes Call Price: {bs_price:.2f} USD")
```

MC Converges to BS Price



Calculated using $S_0 = 330, K = 350, T = 0.25, r = 4\%, \sigma = 25\%$.

Summary of Numerical Methods / Simulations



Running simulations is “slower” than plugging into the analytic formula under Black-Scholes.

If you run many, many simulations, the Monte-Carlo price eventually converges to the true price under Black-Scholes.

Put Prices Under Different Models (**)

Model	Exercise	Price (USD)	Method
GBM	European	30.81	Black-Scholes
GBM	American	31.88	Simulation
Jump-Diffusion	American	35.92	Simulation
Stochastic Vol	American	32.49	Simulation

Additional parameters:

Jump-Diffusion (Merton): $\lambda = 0.75$ (jumps/year), with average jump size $\mu_J = -0.10$

Stochastic Vol (2 state Markov): $\sigma_H = 0.45, \sigma_L = 0.15$, with a 2% transition probability from the low to the high state, and 5% from H to L.



Loose Ends

Futures

The fair delivery price of a forward/futures contract is given by

$$F_0 = S_0 e^{(r+u-q)T}$$

- u is a storage cost as a constant proportion of the spot price (in contrast to $(S_0 + U)e^{(r-q)T}$ if storage costs were in dollar value).
- q is the dividend yield or leasing income on the underlying

Futures

Often, commodities face market disruptions that affect supply/demand, and trading of the commodity. Given a forward delivery price F_0, we back-out the convenience yield using

$$F_0 = S_0 e^{(r+u-q-y)T}$$

where y is the convenience yield on the asset.

The convenience yield measures the extent to which holding the physical asset today is valuable due to concerns over scarcity of a physical commodity, or the extent to which the spot price is overvalued.

Futures

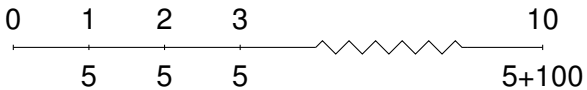
$$F_0 = S_0 e^{(r+u-q-y)T}$$

In the normal case, when the convenience yield is very low, $F_0 > S_0$ and the forward delivery price increases as T increases. We call this contango.

Sometimes, when the spot price is excessively high, $F_0 < S_0$, and the forward delivery price *decreases* as T increases. We call this backwardation.

Bonds: Clean and Dirty Prices

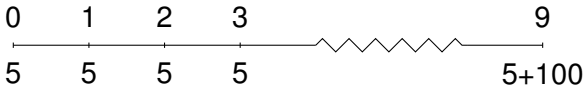
Consider a 10-year coupon bond with a face value of 100, a coupon rate of 5%, annual coupon payments and a YTM of 6%.



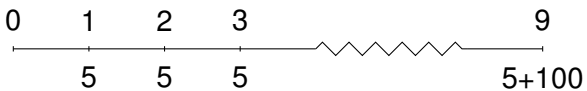
At date 0, you purchase this bond for $p_0 = 92.6399$.

Bonds: Clean and Dirty Prices

Suppose we are 1 year later, immediately before the first coupon is paid,

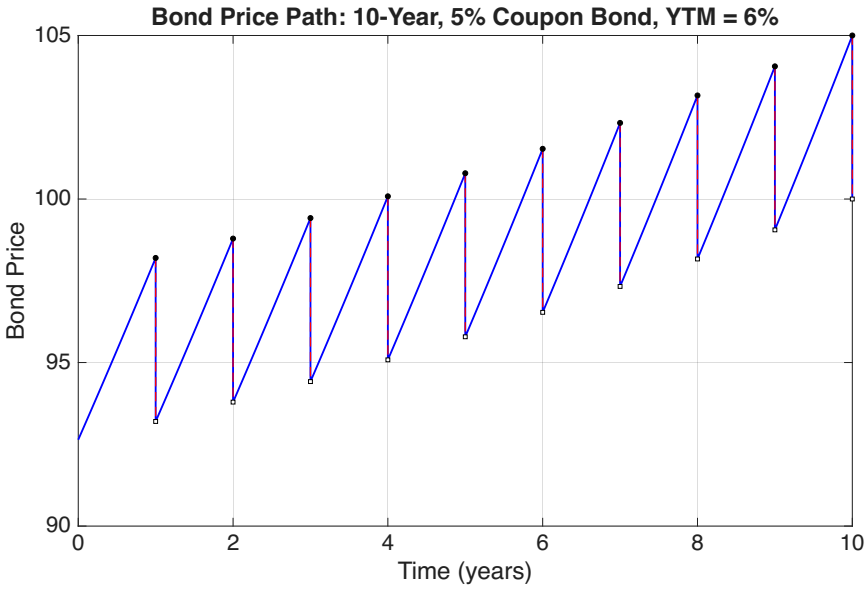


The new price is $p_{1^-} = 98.1983$. Immediately after the coupon is paid, the new price is



$p_1 = 93.1983$.

Bonds: Clean and Dirty Prices



Bonds: Clean and Dirty Prices



Due to time value of money issues, the bond prices would have jumps immediately after each coupon is paid. We refer to this bond price with these sawtooth patterns as the dirty price.

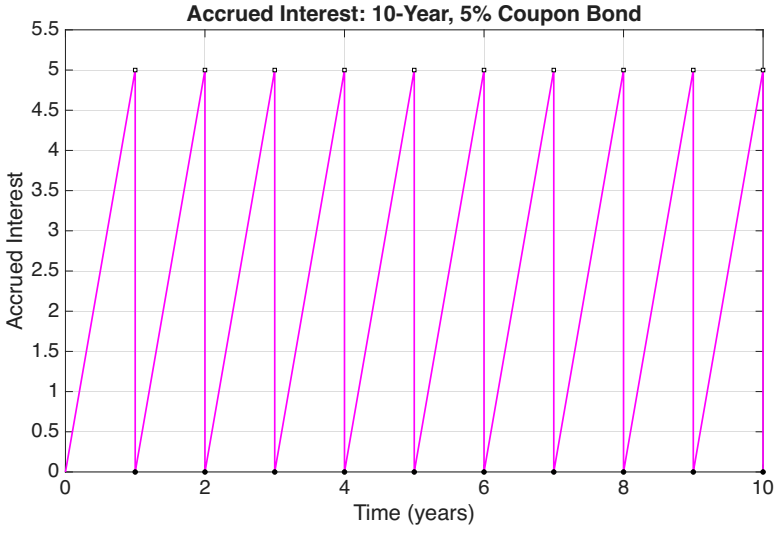
However, traders only care about changes in the bond price due to changes in the yield. Let

$$\text{Accrued Interest} = \text{Coupon amount} \times \left(\frac{\text{Days since last coupon payment}}{\text{Days in current coupon period}} \right)$$

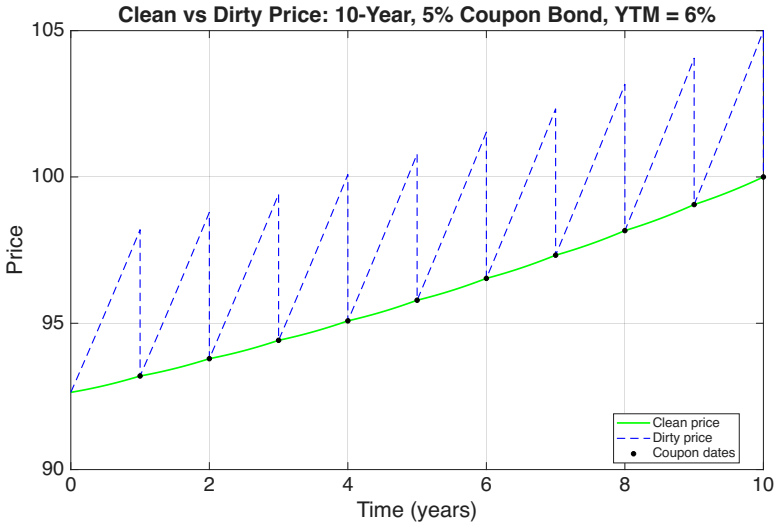
Then,

$$\text{Clean Price} = \text{Dirty Price} - \text{Accrued Interest}$$

Bonds: Clean and Dirty Prices



Bonds: Clean and Dirty Prices



Bonds: Clean and Dirty Prices



The bond price quoted in the market is the clean price.
The actual money that is exchanged is the dirty price.

Accrued interest resets to zero after every coupon payment and linearly increases as the next coupon date approaches.